Symbiosis Institute of Technology, Nagpur

ONE MONTH INTERNSHIP REPORT

for

Bachelor of Technology
in
Computer Science and Engineering

Submitted by

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Semester: V

at

BOTBIRD

Institute Guide

Dr. Smita Singh
Institute Guide, Symbiosis Institute of Technology, Nagpur

August 2026

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Symbiosis International (Deemed University), Pune

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DECLARATION

I, hereby declare that the Industry Internship report submitted herein has been carried out by me in **BOTBIRD** towards partial fulfillment of requirement for the award of Degree of Bachelor of Technology in Computer Science and Engineering. The work is original and has not been submitted earlier as a whole or in part for the award of any degree / diploma at this or any other Institution / University.

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Signature:

Place: Nagpur
Date: August 15, 2026

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CERTIFICATE

This is to certify that the Industry Internship Report, titled **"Cube Portfolio – New Generation Banking Solution,"** has been carried out under our supervision at **BOTBIRD** by **Soumya Bajirao** in partial fulfillment of the requirements for the Bachelor of Technology in Computer Science and Engineering. The work submitted is comprehensive, complete and fit for evaluation.

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Industry Guide

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **BOTBIRD** for providing me with the opportunity to complete my internship in **Software Development** and gain valuable industry experience. This internship has been a highly enriching journey, helping me develop both technical and professional skills.

I am deeply thankful to my mentor/supervisor, **Sanket Sapate**, for his constant guidance, insightful feedback, and encouragement throughout the internship. His expertise and support played a crucial role in enhancing my learning experience. I also extend my appreciation to my colleagues and team members at **BOTBIRD** for their cooperation, assistance, and for making my internship a pleasant and productive experience.

I am immensely grateful to my academic institution, Symbiosis Institute of Technology, Nagpur for facilitating this internship opportunity. I extend my heartfelt appreciation to **Dr. Nitin Rakesh**, Director of Symbiosis Institute of Technology, Nagpur, for his continuous support and encouragement in providing students with industry exposure.

I would also like to extend my deepest gratitude to **Dr. Sagarkumar S. Badhiye**, Head of the Department Computer Science and Engineering, Nagpur, for his constant support, encouragement, and invaluable guidance throughout my learning experience.

A special note of thanks goes to the **Training and Placement Department** for their efforts in providing internship opportunities and career guidance. I would also like to express my sincere gratitude to **Dr. Priya Dasarwar**, Internship In-Charge, for her support in coordinating the internship process and ensuring a smooth learning experience.

Furthermore, I sincerely appreciate the guidance and motivation provided by my Institute Mentor, **Dr. Smita Singh**, whose valuable insights and feedback have greatly contributed to my professional growth during this internship.

Lastly, I would like to acknowledge my family and friends for their unwavering support and encouragement during this period.

Thank you all for your invaluable contributions to my learning and professional growth.

ABSTRACT

This report presents the work carried out during a one-month internship at BOTBIRD, undertaken from 21 May 2026 to 30 June 2026, in partial fulfillment of the requirements for the Bachelor of Technology degree in Computer Science and Engineering. The internship centered on the project *Cube Portfolio – New Generation Banking Solution*, which involved gaining practical exposure to the design and working of a modern, technology-driven banking system.

The internship began with a study of traditional and digital banking processes, followed by research into new-generation digital banking platforms and banking management software such as CRED, PhonePe, Google Pay, Fi Money, and Jupiter, which were examined for comparison and reference purposes only. Building on this understanding, the internship work proceeded to the practical study, installation, and configuration of a MERN-based (MongoDB, Express.js, React.js, Node.js) e-banking application, namely the Sprints-MERN-E-Banking-System project.

The work carried out included setting up the project environment, configuring MongoDB Atlas for database connectivity, running the React.js and Vite-based frontend alongside the Node.js and Express.js backend, and verifying communication between them. Backend route groups covering user management, administrative management, account operations, and account requests were studied in detail. Particular attention was given to the administrative and employee-level workflow, including authentication, role-based access control, and the Owner account creation process, which was tested using the `POST /api/admins/owner/create` endpoint. REST API testing was carried out extensively using Thunder Client to validate request handling, JSON payloads, and protected routes.

During the course of the internship, practical issues such as an undefined controller function, MongoDB Atlas connectivity errors, malformed JSON requests, and authentication requirements on protected routes were encountered and resolved, providing valuable hands-on troubleshooting experience. Through this process, the frontend was successfully executed, the backend was successfully executed, database connectivity was established, and the overall user-level and employee-level workflows of the banking application were analyzed and understood.

Overall, the internship provided practical exposure to full-stack MERN application

architecture, database connectivity, REST API testing, authentication mechanisms, and systematic debugging and documentation practices. It helped connect academic concepts of software engineering and web development with the practical study and operation of a real-world, full-stack banking application.

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1. INTRODUCTION TO COMPANY

1.1 About the Company

BOTBIRD is the organization at which the one-month internship, from 21 May 2026 to 30 June 2026, was undertaken. During the internship, exposure was provided to modern, technology-driven software development practices, with the internship work centered on the study and practical configuration of a full-stack, MERN-based banking application titled *Cube Portfolio – New Generation Banking Solution*. The internship offered an opportunity to understand how a technology organization approaches the design, development, and testing of digital banking systems, and to gain hands-on experience with the tools and workflows used in a professional software environment.

1.2 Historical Background

Detailed historical records regarding the founding and evolution of BOTBIRD were not part of the material made available during the internship. In the context of the internship, BOTBIRD's role was primarily observed through the project work assigned, which involved the study and configuration of a modern e-banking system, reflecting the organization's engagement with contemporary, technology-driven software solutions.

1.3 Location

Specific details regarding the company's registered address and branch locations were not part of the material provided for this internship. The internship itself was carried out remotely or on-site as arranged with BOTBIRD, in accordance with the internship terms communicated by the Industry Guide.

1.4 Operational Structure

The internship work was organized around a single, well-defined project, *Cube Portfolio – New Generation Banking Solution*, under the supervision of the Industry Guide, Sanket Sapate. Work activities progressed from initial research and requirement understanding to

hands-on project setup, configuration, and testing. Tasks were structured in a phase-wise manner: banking-domain research, study of the MERN project architecture, environment installation and configuration, backend and API study, administrative workflow testing, and documentation of results and issues encountered. This structure allowed the internship activities to be closely aligned with the stages of a real software project lifecycle.

1.5 Vision and Mission

Specific, officially published vision and mission statements for BOTBIRD were not part of the material available during the internship and are therefore not reproduced here. Based on the nature of the internship project, BOTBIRD's work reflects a broader commitment within the technology sector to building modern, efficient, and secure digital solutions, such as new-generation banking platforms that streamline customer and administrative workflows through full-stack web technologies.

1.6 Products or Software Solutions

The internship provided exposure to BOTBIRD's involvement in modern software solutions through practical work on the *Cube Portfolio – New Generation Banking Solution* project, a MERN-based (MongoDB, Express.js, React.js, Node.js) e-banking application. This project encompassed both customer-facing banking operations and administrative and employee-level management functions, including user registration and login, account requests, account management, and role-based administrative control. The internship work directly connected to this solution through the study, installation, configuration, and testing of the application's frontend, backend, database, and REST APIs.

2. PROJECT WORK / CASE STUDY

2.1 Introduction

The internship project, titled *Cube Portfolio – New Generation Banking Solution*, involved research into modern banking systems along with the practical study and configuration of a MERN-based (MongoDB, Express.js, React.js, Node.js) e-banking application. The project work included understanding traditional and digital banking, researching new-generation digital banking platforms, and studying banking management software. It also involved analyzing customer and user workflows and employee and administrative workflows, studying the MERN architecture used in the project, installing and configuring the application, setting up MongoDB Atlas, running the frontend and backend, testing REST APIs using Thunder Client, troubleshooting configuration issues, and documenting the process and outcomes.

It is important to note that the internship involved studying, analyzing, configuring, testing, executing, and documenting an existing project rather than developing the entire application from scratch. Through this hands-on process, practical exposure was gained to the working of a full-stack banking application and the technologies underlying it.

2.2 Problem Identification

Modern banking software must provide reliable digital access to banking services while organizing both customer and administrative workflows in a secure and efficient manner. This requires well-defined account management processes, structured handling of account requests, robust authentication and role-based access control, a reliable database-backed system for banking operations, and seamless communication between the frontend and backend components of the application [1].

The internship work focused on understanding how these requirements are represented and implemented in a MERN-based e-banking application. This included examining how user and administrative workflows were structured, how requests for new accounts were routed and processed, and how the system enforced authentication and role-based permissions between regular users and administrators. It should be noted that the internship work involved studying and testing these mechanisms within the scope of the given project, and

does not claim to solve real-world banking problems at a production scale. As shown in

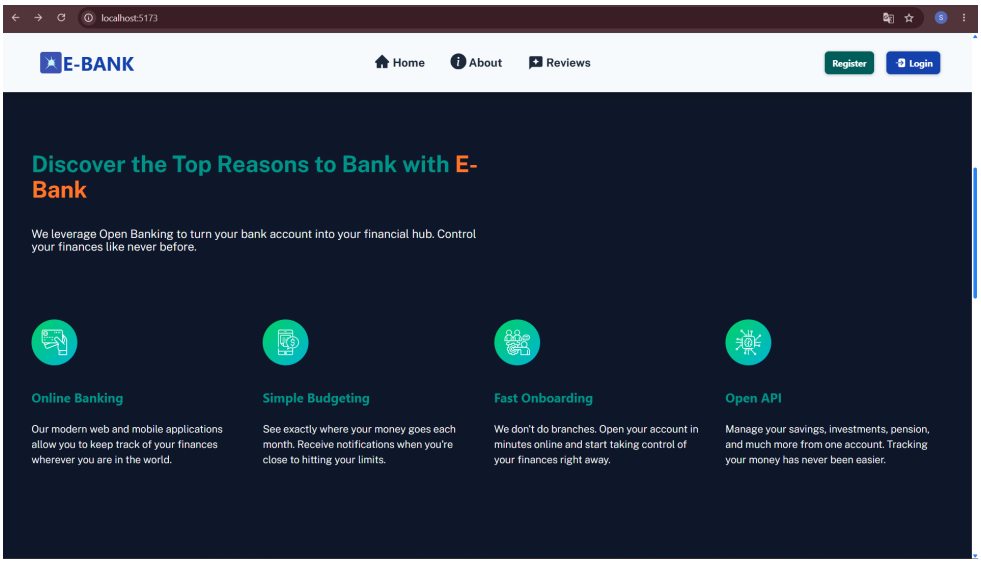


Figure 2.1: Home Page of the E-Bank Application

Figure 2.1, the system interface allows users to login and access various modules.

2.3 Objective

The objectives of the internship were as follows [2]:

1. To understand the working and workflow of modern banking systems.
2. To study new-generation digital banking platforms and banking management software.
3. To analyze customer-level and employee-level workflows in a banking application.
4. To gain practical exposure to MERN-based application architecture, database connectivity, and REST API testing.
5. To improve practical troubleshooting, analysis, and documentation skills through project setup and testing.

2.4 Work Carried Out

The work carried out during the internship progressed through the following stages.

A. Banking System Research

Traditional banking processes and modern digital banking were studied, along with research into new-generation banking platforms and banking management software. Customer onboarding and both user-level and employee-level workflows were also studied.

Platforms such as CRED, PhonePe, Google Pay, Fi Money, and Jupiter were studied as research and comparison examples; it should be noted that these platforms were not developed as part of this internship.

B. MERN Project Study

The project studied was Sprints-MERN-E-Banking-System, structured into a Backend and a Frontend. The technologies used included React.js, Node.js, Express.js, MongoDB Atlas, JavaScript, Vite, Thunder Client, Visual Studio Code, npm, and Git/GitHub.

C. Project Installation and Configuration

This stage involved installing project dependencies, configuring environment variables, and setting up MongoDB Atlas, followed by running the backend and frontend and verifying communication between them.

- Backend started using: `nodemon Backend/server.js`
- Frontend started using: `cd Frontend && npm run dev`
- Both run together using: `npm run both`
- Frontend accessible at: `http://localhost:5173/`
- Backend accessible at: `localhost:5000`

D. Backend/API Study

The following route groups were studied:

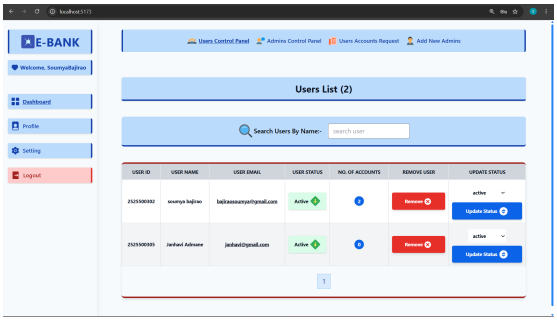
- `/api/users`
- `/api/admins`
- `/api/account`
- `/api/request`

These routes were examined to understand the application's user management, administrative management, account operations, and account request operations.

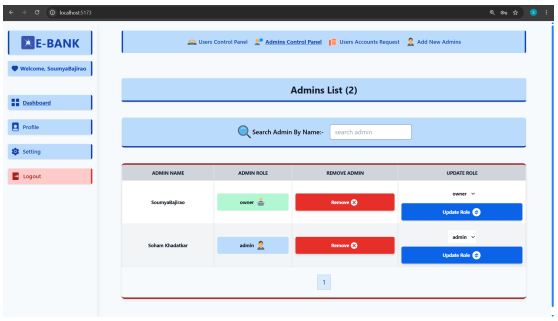
E. Admin/Employee-Level Work

This stage involved studying admin login, authentication, role checking, user management, account request management, account management, owner/admin operations, and role-based access. The route `POST /api/admins/owner/create` was tested for initial Owner account creation.

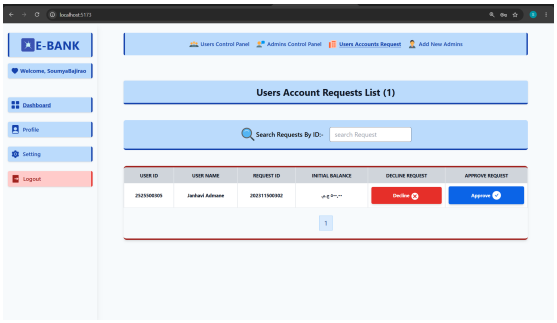
As shown in Figure 2.2, the administrative panel provides controls for managing users, managing admins, reviewing account requests, and adding new administrators.



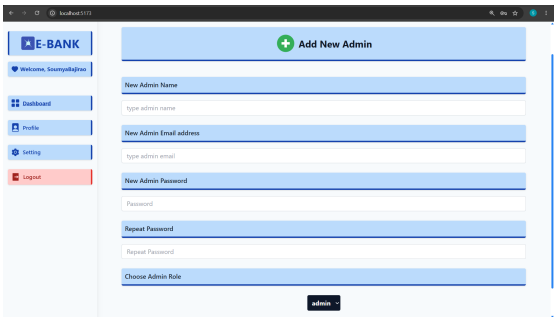
(a) Users Control Panel



(b) Admins Control Panel



(c) User Account Requests



(d) Add New Admin

Figure 2.2: Administrative Panel Screens Studied During the Internship

F. API Testing

Thunder Client was used for sending POST, GET, PUT, and DELETE requests, testing JSON request bodies, checking API responses, testing authentication on protected routes, and testing Owner account creation.

G. Troubleshooting

The following issues were encountered and resolved during the internship:

1. The function `createFirstAdmin` was initially undefined, producing the error:

   ```
   Route.post() requires a callback function but got a
   [object Undefined]
   ```

   The controller import and export statements were checked and corrected, after which `createFirstAdmin` resolved to a function and the backend started successfully.

2. A MongoDB Atlas connectivity issue produced the error:

   ```
   MongooseServerSelectionError: Could not connect to any
   servers in your MongoDB Atlas cluster.
   ```

   This was traced to MongoDB Atlas network/IP access settings, which were investigated and corrected.

3. An invalid JSON request format caused a JSON parsing error during API testing; the request body was corrected.

4. It was identified during API testing that protected administrative routes required authentication, which was addressed while testing those routes.

These issues were encountered and resolved as part of normal development and testing activity during project setup and configuration.

2.5 Solution Provided

The solution studied and configured during the internship was a MERN-based e-banking management application, structured according to the following layered architecture:

React.js + Vite Frontend
↓
REST API
↓
Node.js + Express.js Backend
↓
Middleware + Controllers
↓
MongoDB Atlas

React.js formed the frontend user interface, while **Vite** served as the frontend development and build environment. **Node.js** provided the backend runtime, with **Express.js** handling REST API routing. **Middleware** components managed authentication, authorization, validation, and request processing, while **Controllers** implemented the application logic. **MongoDB Atlas** was used for database storage, and **Thunder Client** was used for API testing.

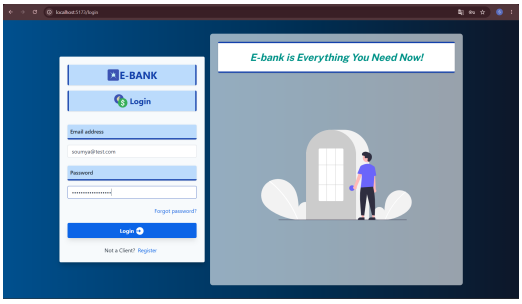
Two major workflows were studied within the application:

User/Customer Level:

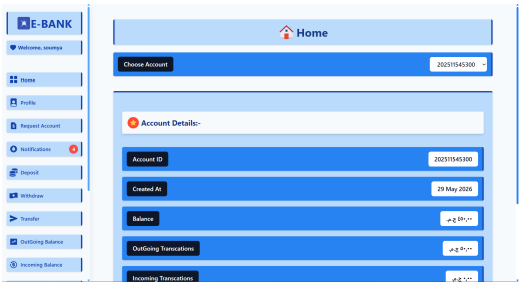
Registration → Login → Account Request → Request Processing → Account Creation → Banking Operations → Deposit / Withdrawal / Transfer → Notifications

Employee/Admin Level:

Admin Login → Authentication → Role Verification → User Management → Account Request Management → Account Management → Administrative Operations



(a) Login Interface



(b) User Dashboard - Account Details

Figure 2.3: Login and User Dashboard Screens of the Application

These workflows reflect the features actually present in the studied project; no additional features beyond those observed in the application have been described here.

As shown in Figure 2.3, users log in through the login interface and are then able to view their account details on the dashboard.

The tools and software used during the internship, along with the system requirements for running the project, are summarized below.

Table 2.1: Software Tools Used in the Project

Sr. No	Tool/Software	Purpose
1	React.js	Frontend Development
2	Node.js	Backend Runtime
3	Express.js	REST API Development
4	MongoDB Atlas	Database Management
5	JavaScript	Application Development
6	Vite	Frontend Development Server
7	Thunder Client	API Testing
8	VS Code	Code Editing and Project Setup
9	GitHub	Version Control
10	npm	Package Management

Table 2.2: System Requirements

Component	Specification
Processor	Intel i5 or above
RAM	8 GB or above
Operating System	Windows 10 / Windows 11
Development Environment	Visual Studio Code
Runtime	Node.js and npm
Database	MongoDB Atlas
API Testing	Thunder Client
Internet	Required for MongoDB Atlas connectivity

2.6 Results

Through the course of the internship, the project dependencies were installed successfully, and both the frontend and backend of the application were executed without errors. MongoDB Atlas connectivity was successfully established during execution, allowing the backend to interact with the database. The application's REST APIs were studied and tested, including the Owner account creation API, using Thunder Client as the testing tool. User-level and employee-level workflows were analyzed to understand how customer and administrative operations were structured within the application. Configuration and debugging issues encountered during setup and testing, including the controller import issue, MongoDB Atlas connectivity error, malformed JSON request, and authentication requirements on protected routes, were investigated and resolved. Overall, the MERN architecture of the project, along with the interaction between the frontend, backend, and database, was understood through this hands-on process. The outcomes of the key testing activities are summarized in Table 2.3.

Table 2.3: Performance Evaluation

Test Case	Description	Result
TC1	MongoDB Atlas Database Connection	Pass
TC2	Frontend Startup using Vite	Pass
TC3	Backend Startup using Node.js and Express.js	Pass
TC4	Owner Account Creation API Testing	Pass
TC5	REST API Testing using Thunder Client	Pass
TC6	User and Employee Workflow Analysis	Completed
TC7	Frontend and Backend Local Execution	Pass

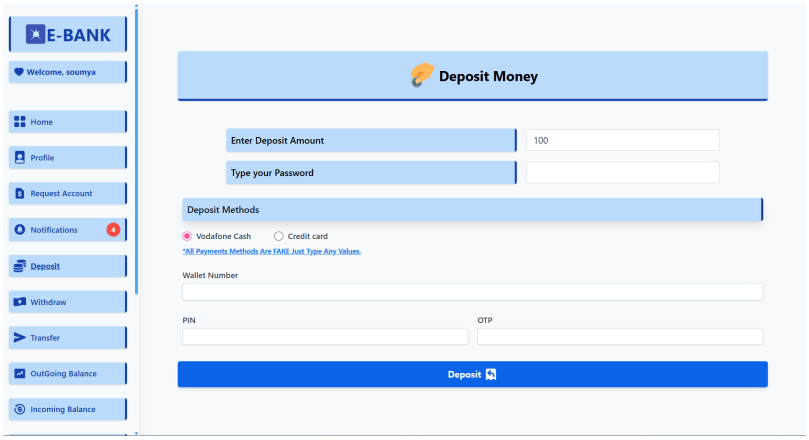


Figure 2.4: Deposit Money Interface of the Application

Figure 2.4 shows the deposit interface, one of the core banking operations available on the user-level dashboard that was studied and tested during the internship.

2.7 Conclusion

The internship at BOTBIRD provided practical exposure to modern banking systems, digital banking, and banking workflows through the study and configuration of the *Cube Portfolio – New Generation Banking Solution* project. Hands-on experience was gained with the MERN architecture, including React.js, Node.js, Express.js, and MongoDB Atlas, along with REST APIs, authentication, and role-based access control. The internship also strengthened practical skills in API testing, troubleshooting, and technical documentation. Overall, the internship helped connect academic concepts of software engineering with practical, hands-on software-system analysis and full-stack application operation.

BIBLIOGRAPHY

[1] J. Smith, “Natural language processing techniques,” *IEEE Transactions on AI*, 2023.

[2] D. Lee, “Ai chatbot development,” in *IEEE International Conference on AI*, 2022.

A. PLAGIARISM REPORT

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A Machine Learning-Based Predictive Model for Early Detection of Breast Cancer

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***Abstract*—Breast cancer is one of the biggest health risks today, and early diagnosis is essential for a better prognosis of the disease. Data-driven approaches such as machine learning emerge successful in diagnosis and breast cancer prediction. This section assesses several general classifiers for prediction such as Random forest (RF), Logistic Regressions (LR) etc. algorithms explored on breast cancer datasets. The dataset is a fine needle aspiration (Biopsy) dataset namely Wisconsin Breast Cancer dataset, which was obtained from the UCI Machine Learning (ML) database. Discussions on the comparisons of different algorithms, feature selection methods, and performance measures are also covered.**

***Index Terms*—Breast-cancer, Machine-learning, Classification algorithms, SVM, Random forest, Feature selection, Data mining.**

I. Introduction

Breast-cancer is a serious and common type of cancer that is a great cause of death among women worldwide [1]. World Health Organization specifies it as a leading cause of death, responsible for around the fifteenth percent of all cancer death rates in women [5]. Timely and accurate detection followed by judicious handling of the disease path in the first stages, improves prognosis and survival rates- vastly [1]. While the accuracy, consistency, and efficiency of traditional diagnostic methods, such as mammography, are problematic in the early stages of the disease, non-invasive methods powered by modern technologies usually demonstrate the strengths of all the mentioned indicators [2]. Figure 1 shows the incidence of breast cancer cases in different countries like South America, North America, East Asia and South-Central Asia. Cancer is a broad term that refers to a group of diseases that share a common characteristic: the abnormal proliferation of body cells or cell multiplication so the body cannot regulate it properly. Unpacking that way, cancerous cells depart from the regulated growth and division orders that healthy cells follow and divide and multiply, basically, at an exorbitant rate, which ends up in the formation and enlargement of malignant tumors or neoplasms [14]. There are various types of cancers

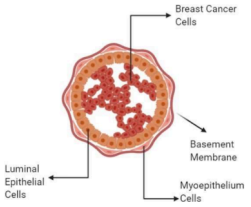


Fig. 1. Cancer Cell

that are named on the basis of the location of concurrence in the human body. For example, lung cancer, breast cancer, colorectal cancer, prostate cancer and skin cancer are the most common type of cancers worldwide. Figure 2 shows some common types of cancers and how common they are based on the number of cases worldwide per year. Cancer was a disease that led to the genome material which was started due to cell division and growth pathway interference. Alterations in genes were the cause of these mutations. These genetic modifications identify the cells for the new job and the specific task, they emphasize controlled cellular growth as well as its checks cells that are healthy contain a complex system of checks and balances which is supposed to govern orderly growth and division; however, in cancer cells, this is disturbed by genetic mutations and leads to uncontrolled growth. In the process, the cells as a whole turn into mutineers, breaching the jurisdiction of usual laws and proliferating endlessly, and eventually become tumors or spread to other

B. AI SIMILARITY REPORT

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	Submitted to Nottingham Trent University	

A Machine Learning-Based Predictive Model for Early Detection of Breast Cancer

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***Abstract*—Breast cancer is one of the biggest health risks today, and early diagnosis is essential for a better prognosis of the disease. Data-driven approaches such as machine learning emerge successful in diagnosis and breast cancer prediction. This section assesses several general classifiers for prediction such as Random forest (RF), Logistic Regressions (LR) etc. algorithms explored on breast cancer datasets. The dataset is a fine needle aspiration (Biopsy) dataset namely Wisconsin Breast Cancer dataset, which was obtained from the UCI Machine Learning (ML) database. Discussions on the comparisons of different algorithms, feature selection methods, and performance measures are also covered.**

***Index Terms*—Breast-cancer, Machine-learning, Classification algorithms, SVM, Random forest, Feature selection, Data mining.**

I. Introduction

Breast-cancer is a serious and common type of cancer that is a great cause of death among women worldwide [1]. World Health Organization specifies it as a leading cause of death, responsible for around the fifteenth percent of all cancer death rates in women [5]. Timely and accurate detection followed by judicious handling of the disease path in the first stages, improves prognosis and survival rates- vastly [1]. While the accuracy, consistency, and efficiency of traditional diagnostic methods, such as mammography, are problematic in the early stages of the disease, non-invasive methods powered by modern technologies usually demonstrate the strengths of all the mentioned indicators [2]. Figure 1 shows the incidence of breast cancer cases in different countries like South America, North America, East Asia and South-Central Asia. Cancer is a broad term that refers to a group of diseases that share a common characteristic: the abnormal proliferation of body cells or cell multiplication so the body cannot regulate it properly. Unpacking that way, cancerous cells depart from the regulated growth and division orders that healthy cells follow and divide and multiply, basically, at an exorbitant rate, which ends up in the formation and enlargement of malignant tumors or neoplasms [14]. There are various types of cancers

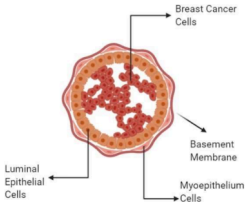
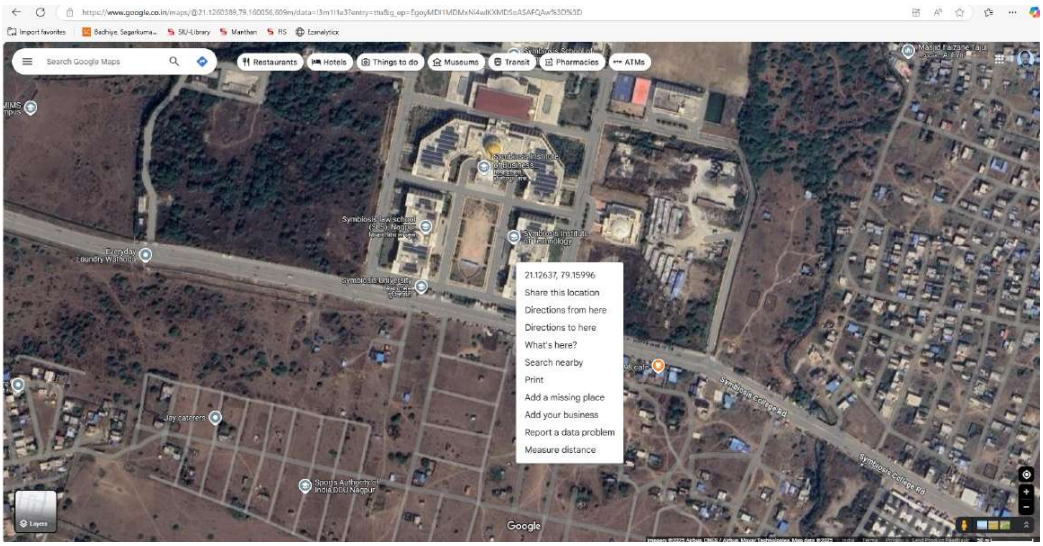
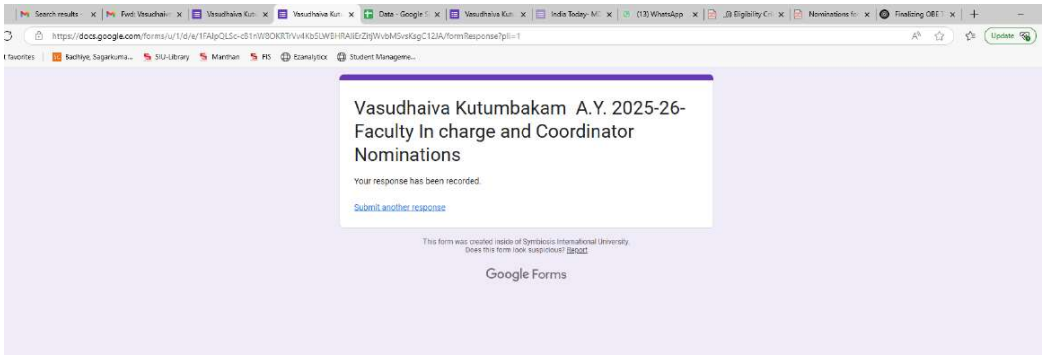


Fig. 1. Cancer Cell

that are named on the basis of the location of concurrence in the human body. For example, lung cancer, breast cancer, colorectal cancer, prostate cancer and skin cancer are the most common type of cancers worldwide. Figure 2 shows some common types of cancers and how common they are based on the number of cases worldwide per year. Cancer was a disease that led to the genome material which was started due to cell division and growth pathway interference. Alterations in genes were the cause of these mutations. These genetic modifications identify the cells for the new job and the specific task, they emphasize controlled cellular growth as well as its checks cells that are healthy contain a complex system of checks and balances which is supposed to govern orderly growth and division; however, in cancer cells, this is disturbed by genetic mutations and leads to uncontrolled growth. In the process, the cells as a whole turn into mutineers, breaching the jurisdiction of usual laws and proliferating endlessly, and eventually become tumors or spread to other

C. PROJECT SCREENSHOTS




Vasudhaiva Kutumbakam A.Y. 2025-26- Faculty In charge and Coordinator Nominations
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